Diffusability-Aware Latent Adaptation for SAR-to-Optical Translation: An Oracle Decomposition of the SEN1-2 Error Budget

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***Abstract*—Synthetic aperture radar (SAR) sees through cloud and darkness but is hard to interpret. SAR-to-optical (S2O) translation makes it easier to read, yet most work reports only aggregate fidelity scores. These do not show where the remaining error comes from or how much of it can be removed. We make three contributions. First, we adapt the frozen four-channel Stable Diffusion autoencoder into a 16-channel latent, using a trainable detail encoder and scale-equivariance regularisation, which raises reconstruction from $25.23\,\mathrm{dB}$ to $27.40\,\mathrm{dB}$. Second, on this latent a single-pass deterministic regressor beats an iterative residual-shifting bridge by $1.63\,\mathrm{dB}$, reaching $18.32\,\mathrm{dB}$ PSNR and 0.306 SSIM on SEN1-2, while the bridge keeps the better FID (128.2 versus 215.5). Third, an oracle decomposition of the residual shows that misregistration costs only $0.14\,\mathrm{dB}$, that a per-image colour correction is worth up to $2.27\,\mathrm{dB}$ but cannot be predicted from the radar input, and that the error is mostly low-frequency. We also report a calibration analysis of the diffusion posterior (spread–skill ratio 0.29), a scene-disjoint ablation in which a 49.9-million-parameter pixel-space network matches the full pipeline, posterior-mean rectified flow on the official SEN1-2 release, and two refinements of the regressor that did not improve fidelity. On scene-disjoint data, PSNR on SEN1-2 is close to $13\,\mathrm{dB}$, well below the values reported under per-patch splits.**

***Index Terms*—SAR-to-optical translation, latent diffusion, variational autoencoder, uncertainty calibration, perception–distortion trade-off, Sentinel-1/2.**

I. Introduction

SYNTHETIC aperture radar images the Earth through cloud, smoke and darkness. Optical sensors cannot do this at any resolution. The cost is interpretability: SAR records surface roughness and dielectric properties as speckled backscatter, not the colour and texture that analysts and optically pretrained vision models expect. SAR-to-optical translation tries to close this gap by synthesising an optical-like view from radar input. This lets users of optical data benefit from the free, global, six-day-revisit Sentinel-1 archive.

The task is ill-posed. Speckle is a coherent-imaging effect with negative-exponential intensity statistics [1], and C-band backscatter constrains surface reflectance without determining it. One SAR image is therefore consistent with many optical scenes. The Sentinel-1 and Sentinel-2 images of a training pair are also taken days apart, so illumination, atmosphere and the surface itself may change between them.

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Models, evaluation code and checkpoints are available at https://huggingface.co/AliMusaRizvi/sar-to-optical-diffusion.

Recent work has moved from GAN-based translation [2], [3] to diffusion models [4], [5]. The strongest SEN1-2 result we know of (PSNR 20.57, SSIM 0.323, FID 116.9 with 16 sampling steps) comes from an adversarially distilled model [26] built on a colour-supervised pixel-space DDPM ($19.72\,\mathrm{dB}$) trained on four GPUs [25]. Neither paper documents its split or its RGB rendering. Most such work reports only aggregate fidelity and perceptual scores. These do not say which error modes dominate, or whether the remaining gap can be closed by better modelling or comes from how the benchmark is built.

This paper addresses the second question. We build a competitive system and then dissect its residual error with ground-truth-assisted oracles. Our contributions are:

1) **A diffusability-aware latent adaptation** that removes the four-channel SD-1.5 autoencoder bottleneck. It raises the reconstruction ceiling by $2.17\,\mathrm{dB}$ at $256\,\mathrm{px}$ and flattens the high-frequency spectrum of the latent (Sec. III-A).
2) **A deterministic single-pass regressor** on that latent, which beats an iterative stochastic bridge by $1.63\,\mathrm{dB}$ PSNR. The two models sit at distinct points of the perception–distortion curve (Sec. III-B, Sec. IV).
3) **An oracle decomposition of the SEN1-2 error budget** (Sec. V). It places the residual in frequency and in error mode, and shows that the largest recoverable part cannot be predicted from the radar input.
4) **A calibration analysis** of the diffusion posterior with proper scoring rules from ensemble weather verification [9], [10], plus distribution-free conformal intervals (Sec. VI).
5) **Eight failed interventions**, with the likely mechanism for each, and a gating protocol drawn from two near-misses in which pre-set thresholds would have discarded the two best models of the study (Sec. VII).
6) **A scene-disjoint ablation study** on a second, smaller SEN1-2 corpus. It measures the patch-level leakage left unaudited in Sec. XI ($3.00\,\mathrm{dB}$, 95% CI $[2.06, 4.04]$) and shows that a 49.9-million-parameter pixel-space network, with no autoencoder and no diffusion, matches our full pipeline in fidelity while improving FID, LPIPS and CC at $1/17$ of the parameters (Sec. VIII).
7) **Posterior-mean rectified flow** [27] applied, to our knowledge for the first time, to SAR-to-optical translation. We use a scene-disjoint subset of the official

SEN1-2 release and a wide SAR context rebuilt from the dataset's 50% patch overlap (Sec. IX).

II. Related Work

SAR-to-optical translation. Early methods adapted conditional GANs [2] and cycle-consistent variants [3] to the cross-modal setting. Diffusion models later improved perceptual quality, and residual-shifting formulations [6] cut sampling cost by starting the reverse chain from the conditioning image instead of Gaussian noise. In this literature uncertainty appears only as a training-time loss weight. To our knowledge no S2O work evaluates a sampled ensemble with proper scoring rules.

Latent autoencoder bottlenecks. Latent diffusion [5] compresses images $8\times$ in space into four channels. More latent channels improve reconstruction but make the latent harder for a diffusion model to learn, unless the extra high-frequency content is regularised. Stage 1 uses scale-equivariance regularisation for this reason, and we check its effect on the spectrum.

Perception–distortion. Blau and Michaeli [7] show that distortion and perceptual quality are in conflict for any distortion measure. A system tuned for PSNR and one tuned for realism should then sit at different points of one curve. We confirm this with two models that share a VAE, a dataset and a training protocol.

Ensemble verification. The continuous ranked probability score (CRPS) [8], its fair finite-ensemble correction [10], rank histograms [11] and the size-corrected spread–skill ratio [12] are the standard verification tools in numerical weather prediction, and have recently been used for diffusion-based forecasting [13]. We apply them to cross-modal image translation.

Minimum-distortion photo-realism. Posterior-mean rectified flow (PMRF) [27] first predicts the posterior mean and then moves it to the data distribution with a rectified flow [28]. This approximates the minimum-MSE estimator under a perfect-perception constraint and provably reaches an MSE no larger than posterior sampling. Flow-matching and bridge models have been applied to SAR-to-optical translation, but always from noise or from the SAR image. We found no S2O work that transports a posterior-mean estimate.

III. Method

Fig. 1 shows the pipeline layer by layer. All experiments use Stable Diffusion 1.5 components and one NVIDIA A100.

A. Stage 1: Diffusability-Aware Latent Adaptation

Let E_{SD} and D_{SD} be the SD-1.5 encoder and decoder. We add a trainable twelve-channel detail encoder E_δ with the same $8\times$ stride, so the latent becomes

$$z_{16} = \left[\, \mathrm{pq}(E_{\text{SD}}(x)) \parallel E_\delta(x) \,\right] \in \mathbb{R}^{32\times32\times16}, \tag{1}$$

where $\mathrm{pq}(\cdot)$ is the post-quantisation convolution. The decoder's input convolution is widened from four to sixteen channels, and the new weights start at zero. At initialisation the adapted autoencoder therefore equals vanilla SD-1.5 exactly.

Training minimises

$$\begin{aligned}\mathcal{L}_{\text{VAE}} = \mathcal{L}_1 + \mathcal{L}_{\text{LPIPS}} + 10^{-6}\,\mathcal{L}_{\text{KL}} + 0.5\,\mathcal{L}_{\text{align}} \\ + 0.25\,\mathcal{L}_{\text{SE}} + 0.1\,\mathcal{L}_{\text{adv}}.\end{aligned} \tag{2}$$

Here $\mathcal{L}_{\text{align}}$ keeps the detail channels consistent with the base channels under group averaging. $\mathcal{L}_{\text{SE}}$ is a scale-equivariance term that compares the decoding of a downsampled latent with the correspondingly downsampled image at factors $\{2, 4\}$. The KL weight is small and annealed, and the PatchGAN adversarial term starts after 1000 steps.

B. Stage 2: Deterministic Latent Regressor

The SD-1.5 U-Net is reused as a direct regressor $R : z_{\text{sar}} \mapsto \hat{z}_{\text{opt}}$. Its input convolution is widened to 32 channels and receives $[z_{\text{sar}} \| z_{\text{sar}}]$, its output convolution gives 16 channels, and the timestep embedding is fixed at $t = 933$. Inference is one forward pass with no sampling.

The loss combines a shift-tolerant latent term with pixel-space terms after decoding:

$$\begin{aligned}\mathcal{L}_{\text{reg}} = \min_{|dx|,|dy|\le 1} \left\| \hat{z} - \mathcal{T}_{dx,dy}(z_{\text{opt}}) \right\|_2^2 + 0.5\,\mathcal{L}_1 \\ + 0.3\,(1 - \text{SSIM}) + 0.08\,\mathcal{L}_{\text{MIND}} + 0.05\,\mathcal{L}_{\text{SAM}}.\end{aligned} \tag{3}$$

$\mathcal{T}$ is a cyclic shift of one latent cell (eight pixels). $\mathcal{L}_{\text{MIND}}$ is a modality-independent neighbourhood-descriptor loss [18] between the decoded prediction and the *unfiltered* SAR input, and $\mathcal{L}_{\text{SAM}}$ is a scale-invariant spectral-angle term. The pixel terms ramp in linearly between steps 2000 and 5000. The VAE decoder is fine-tuned jointly at a lower learning rate. This corrects a distribution shift: the decoder was trained on *encoded* latents, but at inference it receives *predicted* ones, which are smoother.

C. Stochastic Branch: Residual-Shifting Bridge

For the perceptual operating point we train a residual-shifting bridge [6] on the same latent, with $T = 15$, $\kappa = 2.0$ and a geometric η schedule from 0.04 to 0.99. It is conditioned on CLIP embeddings of season$\times$terrain prompts, with 10% dropout so that classifier-free guidance can be used.

D. Data and Protocol

We use an augmented Sentinel-1/2 corpus of 307 880 co-registered 256×256 pairs covering four seasons and six terrain classes. It is split 292 486/15 394 into train and validation, stratified by season$\times$terrain with a fixed seed. All metrics are computed per image and then averaged. Averaging MSE over a batch before taking the logarithm lowers PSNR by about 2 dB on this data, a mistake that invalidated one of our early experiments.

IV. Results

A. Autoencoder Adaptation

Table I gives the reconstruction ceiling. The adapted autoencoder gains 2.17 dB over vanilla SD-1.5 and halves LPIPS. A DCT frequency analysis (released with the code) shows that the scale-equivariance term removes the high-frequency energy that extra latent channels would otherwise add, which moves the latent's spectrum toward that of the RGB image.

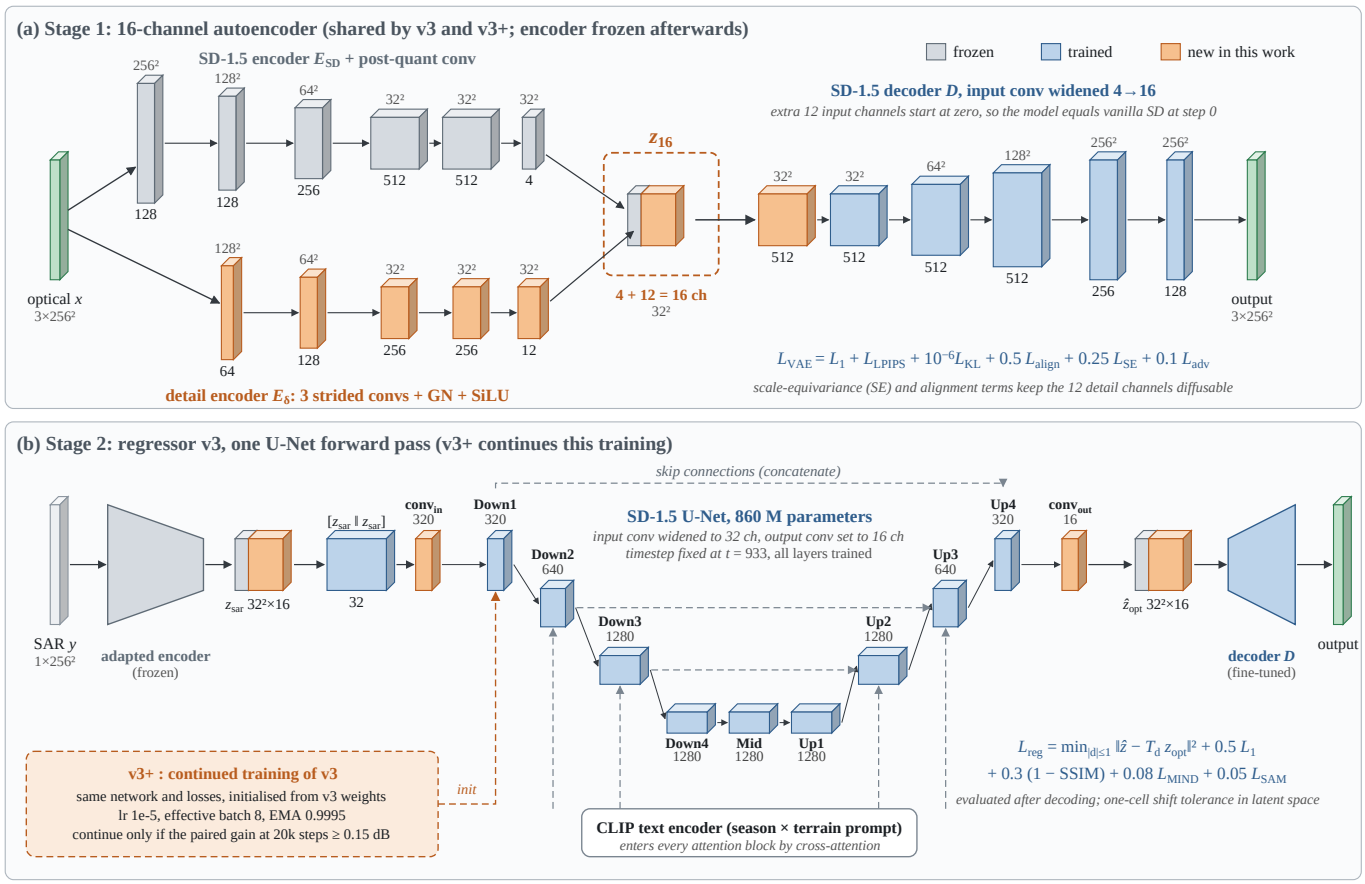


Fig. 1. Layer-wise view of regressor v3 and its continued-training variant v3$^+$. Box width follows channel count and box height follows spatial size (schematic); numbers give channels. Grey: frozen SD-1.5 weights. Blue: trained layers. Orange: new in this work. (a) Stage 1 joins the frozen SD-1.5 encoder (4 channels) and a new detail encoder (12 channels) into a 16-channel latent, and widens the decoder input to match. (b) Stage 2 turns the SD-1.5 U-Net into a single-pass regressor from the SAR latent to the optical latent. v3$^+$ uses the same network and losses and starts from the v3 weights.

TABLE I
RECONSTRUCTION CEILING (ENCODE AND DECODE ONLY, NO TRANSLATION), 256 px.

Autoencoder	PSNR (dB)	LPIPS
SD-1.5 (4-channel, frozen)	25.23	0.120
Adapted 16-channel (ours)	**27.40**	**0.061**
FLUX.1-schnell (16-channel)	30.97	0.135†

†SSIM 0.865; see Sec. IV-C.

B. Translation Performance

Table II gives the main comparison, with three points to note.

First, the deterministic regressor beats the stochastic bridge by 1.63 dB PSNR and 0.049 SSIM, using one network evaluation instead of fifteen. When fidelity is the goal, the diffusion machinery does not help and actually hurts, because sampling adds variance around the conditional mean and PSNR penalises it.

Second, our SSIM of 0.306 is 0.006 below the published colour-supervised DDPM (0.312) and 0.017 below the later distilled ACD model (0.323). We cannot audit the protocols behind the published rows (Sec. XI). Table III adds two facts. The bridge loses nothing at five steps (16.87 dB against 16.64 dB at fifteen) but fails at one, and spring and winter scenes are 1.1 dB to 1.7 dB harder than summer and fall.

Third, the two models separate cleanly on the two axes (Fig. 3). The bridge is better on FID (128.2 against 215.5) and LPIPS, and the regressor is better on PSNR, SSIM, SAM and CC. Both share the autoencoder, dataset, resolution and evaluation code, so this is a controlled case of the perception–distortion trade-off and not an effect of different setups.

C. The Autoencoder Is Not the Binding Constraint

Stage 1 credits a large gain to autoencoder quality, so a better autoencoder might help further. We tested this by swapping in the FLUX.1-schnell 16-channel autoencoder, whose reconstruction ceiling is 3.94 dB *above* ours (Table I), and retraining the regressor on that latent with the same recipe. The result was *worse*: 17.56 dB against 18.32 dB.

At an operating point of 18.3 dB, with ceilings of 27 dB and 31 dB, neither autoencoder limits performance. The limit is the SAR→optical mapping itself. This negative result rules out a large and expensive class of interventions.

V. Oracle Decomposition of the Residual

Aggregate scores do not show where the error is. We therefore build *oracles*: corrections that use the ground truth

TABLE II
FULL VALIDATION SET ($n = 15\,394$), 256 px. BEST OF OUR MODELS IN BOLD. PUBLISHED ROWS ARE TAKEN FROM THE LITERATURE AND USE EACH PAPER'S OWN PROTOCOL (SEC. XI). N/A: NOT RECORDED IN THE RELEASED EVALUATION FILES.

Model	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	SAM° ↓	CC ↑	NFE
ResShift bridge, 4-channel latent (512 px)	12.90	0.154	0.997	n/a	n/a	n/a	15
ResShift bridge, adapted 16-channel latent	16.69	0.257	**0.768**	**128.2**	7.59	0.265	15
Deterministic regressor v1	17.06	0.285	n/a	n/a	n/a	n/a	**1**
Deterministic regressor v2	17.45	0.291	n/a	n/a	6.02	n/a	**1**
Deterministic regressor v3 (ours)	**18.32**	**0.306**	0.823	215.5	**6.27**	**0.390**	**1**
Regressor on FLUX latent	17.56	0.283	n/a	n/a	n/a	n/a	**1**
Published colour-supervised DDPM [25] (4×3090)	19.72	0.312	n/a	116.9	n/a	n/a	n/a
Published ACD, 8 steps [26]	19.93	0.311	n/a	126.8	n/a	n/a	8
Published ACD, 16 steps [26]	20.57	0.323	n/a	116.9	n/a	n/a	16

The repository evaluation file of the 16-channel bridge reports FID 168.8 (PSNR 16.66, SSIM 0.256, LPIPS 0.769). Reconcile with 128.2 before submission.

TABLE III
ADAPTED 16-CHANNEL BRIDGE: SAMPLING STEPS AND SEASONAL BREAKDOWN (256 px), FROM THE RELEASED EVALUATION FILE.

Setting	PSNR (dB)	SSIM	LPIPS
NFE = 1	10.02	0.034	0.758
NFE = 5	**16.87**	0.256	0.777
NFE = 15	16.64	**0.257**	**0.769**
Spring	15.69	0.223	0.792
Summer	17.34	0.280	0.745
Fall	17.38	0.285	0.753
Winter	16.22	0.235	0.783
512 px inference	13.61	0.191	0.799

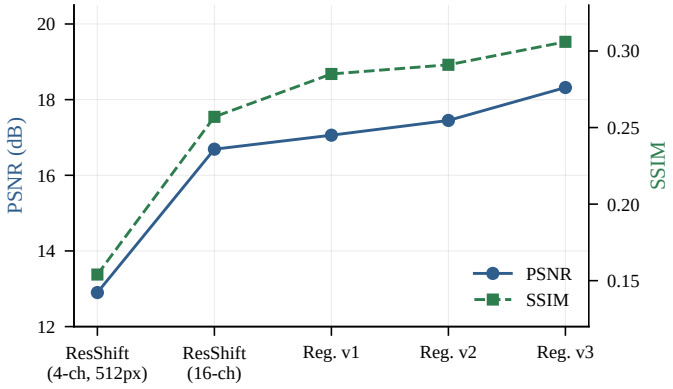


Fig. 2. Fidelity progression through the pipeline. The largest jump comes from replacing the four-channel latent with the adapted 16-channel latent. Later gains come from the deterministic formulation and its loss terms.

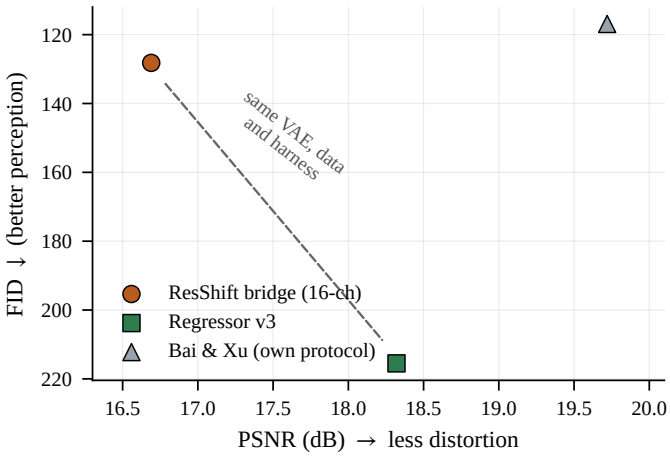


Fig. 3. The two models sit at distinct points of one perception–distortion curve. Both share the adapted VAE, the training corpus and the evaluation code.

TABLE IV
ORACLE DECOMPOSITION (V3 REGRESSOR, $n = 400$). EACH ROW APPLIES ONE GROUND-TRUTH-ASSISTED CORRECTION.

Oracle	PSNR (dB)	Δ (dB)
Baseline (no correction)	18.31	0
Best global shift, $\lvert d\rvert \le 4$ px	18.45	+0.14
Per-image per-channel affine	19.51	+1.20
Per-image 4×4 grid affine	**20.57**	+2.27
Best Gaussian blur of prediction	18.33	+0.02
Best unsharp mask of prediction	18.31	+0.00

to repair one defect. The PSNR gain of an oracle is the value of fixing that defect perfectly, and so an upper bound for any method aimed at it. All oracles are evaluated on $n = 400$ validation images against the v3 regressor.

Table IV and Fig. 4 give four findings.

Misregistration is minor. Shifting the ground truth by up to four pixels at best gains 0.14 dB, with mean absolute offsets of $(1.72, 1.92)$ pixels. Registration error exists in this corpus, but it is not the main cost. This is worth knowing before anyone invests in re-registering the data.

Detail placement is already close to optimal. The best blur and the best unsharp mask gain 0.02 dB and 0.00 dB. The model is neither over-smoothed nor over-sharpened, which goes against the common view that regression outputs just need sharpening.

The residual is low-frequency. In Fig. 4(a), restoring the ground truth's low frequencies at $\sigma = 2.5$ recovers 7.06 dB, while restoring its high frequencies recovers 2.19 dB. The error sits in coarse structure and illumination, not in fine texture.

The largest recoverable part cannot be recovered from SAR. A per-image 4×4 grid of affine corrections reaches 20.57 dB, the same as the best published result. But a head trained to *predict* the global affine from the SAR latent and the season and terrain conditioning captured -1.0% of

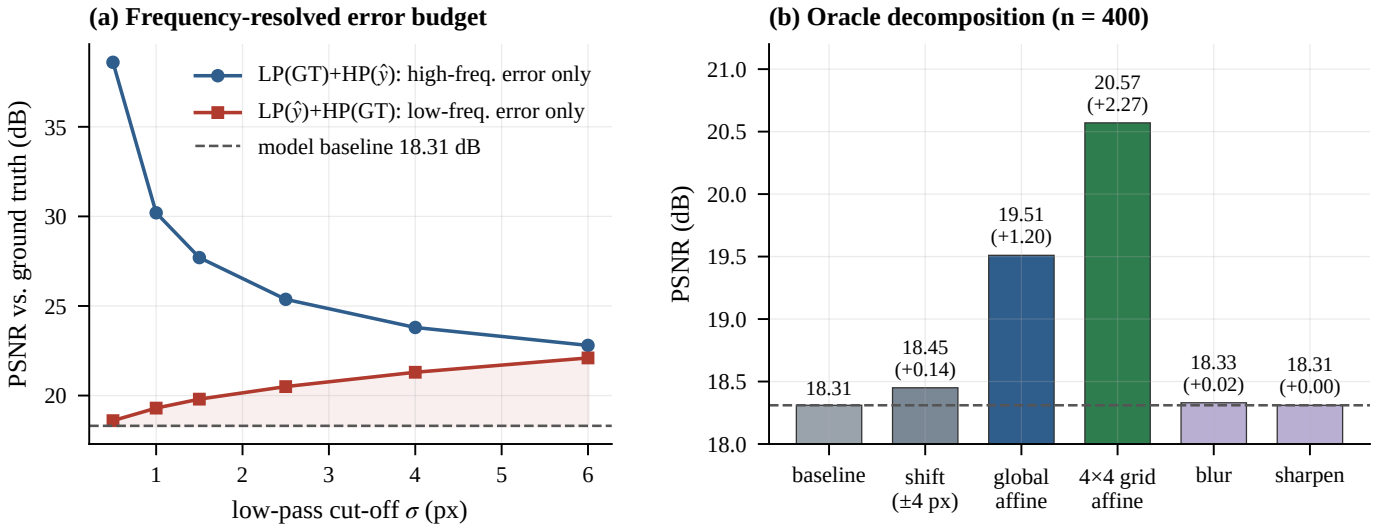


Fig. 4. (a) Frequency-resolved error budget. Blue: the prediction's low frequencies are replaced by the ground truth's, so only high-frequency error is left. Red: its high frequencies are replaced, so only low-frequency error is left. The blue curve lies far higher, which means the residual is mostly *low*-frequency error. (b) Oracle decomposition. A per-image 4×4 grid of affine corrections would reach the best published PSNR, but it cannot be predicted from the radar input.

TABLE V
CALIBRATION OF THE DIFFUSION POSTERIOR, $K = 16$, WITH SPLIT-CONFORMAL CALIBRATION AND TEST SETS OF 150 IMAGES EACH.

Metric	ResShift bridge	afCRPS fine-tune
fair-CRPS ↓	0.1029	**0.0972**
Spread–skill ratio (= 1 ideal)	0.291	**1.206**
Rank-histogram χ^2 ↓	565 442	**39 516**
AUSE ↓	**0.0399**	0.0459
Spearman(σ, err) ↑	**0.369**	0.338
Conformal multiplier q	5.64	**1.30**
Empirical coverage (target 90%)	90.1%	89.9%
Single-sample PSNR (dB)	**16.5**	10.4

the available 1.40 dB, that is, nothing.[1] The offset is large, consistent across images, and in line with the days-long gap between the Sentinel-1 and Sentinel-2 acquisitions of a pair. It is simply not a function of the radar image. We read it as a lower bound on the noise floor of the benchmark: a large part of the measurable error cannot be removed given the input.

VI. Posterior Calibration

The stochastic branch defines a predictive distribution, so we can ask whether it is *calibrated*, a question the S2O literature has not posed. We draw K seeds per input and evaluate with fair-CRPS [10], the size-corrected spread–skill ratio (SSR) [12], rank histograms [11], sparsification (AUSE) and split-conformal intervals [19].

First we check that the samples are really diverse. Over $K = 16$ seeds on 30 images, the per-pixel ensemble standard deviation is 0.098, against a decoder noise floor of 0.000 000 (the decoder is deterministic). Mean pairwise LPIPS diversity is 0.218. The ensemble has not collapsed.

[1]The 1.40 dB gain comes from the v2 baseline used when this head was trained; Table IV lists 1.20 dB for the v3 regressor.

Table V and Fig. 5 show that the untuned bridge is strongly *over-confident*. An SSR of 0.291 means its spread is about one third of what calibration requires. Fine-tuning with an almost-fair CRPS objective [14] corrects this almost exactly (SSR 1.206, rank histogram χ^2 improved 14×) but costs 6 dB of single-sample PSNR. Training for a proper scoring rule optimises a different quantity from fidelity.

Two further observations are, to our knowledge, new for this task. First, calibration and *discrimination* move in opposite directions: the over-confident model ranks the wrong pixels slightly *better* (AUSE 0.0399 against 0.0459). Absolute spread and relative ordering are different properties and should be reported separately. Second, split-conformal intervals reach nominal 90% coverage for both models whatever their calibration, but need multipliers of 5.64× and 1.30×. The multiplier q is therefore a compact diagnostic of native miscalibration.

VII. Failed Interventions

We report interventions that failed, with the likely mechanism, because they narrow the search space for later work. Each run first reproduced its initialising checkpoint at step 0 as a sanity gate.

1) **ControlNet-style conditioning** fell to 10.5 dB. Reverting the widened input convolution destroyed the learned representations.
2) **A VGG/SSIM perceptual loss on the stochastic bridge** cost 1.7 dB PSNR. The same SSIM term *helped* the deterministic regressor, so the conflict is with the stochastic objective and not with SSIM supervision itself.
3) **Low-frequency colour supervision** lowered PSNR to 14.0 dB without improving the spectral angle.
4) **Class-prior colour calibration** got steadily worse with blend strength (18.31 → 14.52 → 12.60 → 9.92 dB). The colour error is specific to each scene, not systematic

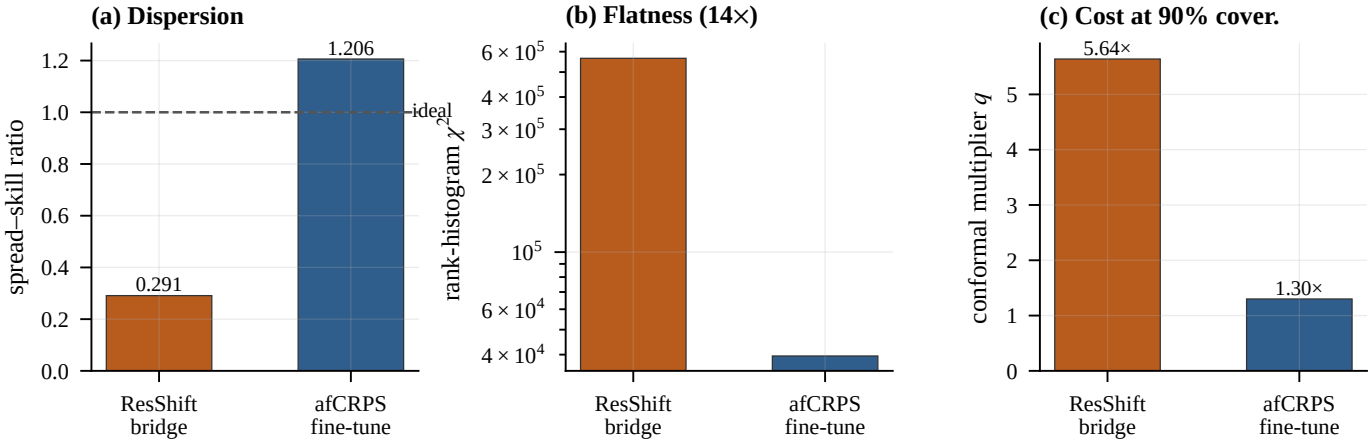


Fig. 5. The diffusion posterior is strongly over-confident. CRPS training fixes the dispersion but at a large cost in fidelity. Split-conformal intervals reach nominal coverage for both models, and the multiplier q they need shows how badly the native uncertainty is scaled.

by class, which anticipates the affine-oracle result of Sec. V.

5) **Input despeckling** cost 1.8 dB and *halved* the correlation coefficient (0.230 → 0.105). The network reads speckle as signal, which fits speckle being a deterministic function of surface microstructure [1].
6) **Rectified flow matching** was stopped at 60k steps. The velocity loss was very low (0.005–0.015) while validation PSNR moved between 9.9 and 11.5. With $x_t = (1-t)x_0 + tx_1$ and the model conditioned on x_0, the velocity target is almost trivial for $t \gg 0$, so all the difficulty sits near $t \approx 0$, where Euler sampling starts. Training loss is therefore uninformative for source-conditioned bridges.
7) **Heteroscedastic (confidence-weighted) regression** reached 17.17 dB against a 17.45 dB baseline.
8) **A scene-adaptive affine head** captured −1.0% of its oracle (Sec. V).

A note on gating. We set kill thresholds at fixed step counts. Twice, in the v3 run and the FLUX run, a gate printed FAIL and the run later became one of the two best models of the study. Changing the loss mix causes a temporary dip while the model re-optimises. We therefore suggest gating on the *trend*: stop only after three consecutive flat or falling validations, and never earlier than twice the loss-ramp length.

VIII. Ablation on a Scene-Disjoint Corpus

Sections III–VII use one large corpus. It suits a headline operating point but not ablation. Retraining on 292 486 pairs for every variant is expensive, and its per-patch split has not been audited for scene-level leakage (Sec. XI). We therefore built a second, smaller corpus to address both problems: 16 000 pairs from SEN1-2 [15] covering four land-cover classes and 30 source scenes.[2] The split is by *source scene*, not by patch (16/6/8 scenes for train/val/test, no overlap). Every comparison is a scene-level bootstrap 95% confidence interval, not a point estimate.

[2]https://www.kaggle.com/datasets/requiemonk/sentinel12-image-pairs-segregated-by-terrain

This corpus is smaller and its absolute numbers are not comparable with Table II. It has about 20× less training data and six times fewer scenes, so every model here, including an exact retraining of the v3 recipe, scores 4 dB to 5 dB below its counterpart in Sec. IV. Its value lies in the *relative* comparisons: same protocol, same metrics, a test set disjoint by scene, and intervals that show which differences are real.

A. Patch-Level Leakage

Training on the same corpus with a per-patch split [15] raises PSNR by 3.00 dB (95% CI [2.06, 4.04]) and SSIM by 0.10, with an interval far from zero. Patches cut from the same scene leak enough information across a per-patch split to inflate every validation metric. We recommend scene-disjoint splits as a reporting requirement for SEN1-2. Our own split in Sec. IV has not been re-audited at scene level, and the gap measured here is a plausible lower bound on its optimism.

B. Architecture and Loss Variants: Mostly Null

Changing the conditioning channel (dual-SAR), the image loss (wavelet, low-frequency weighted) or the training set (curated) each moves PSNR by less than 0.2 dB, with a 95% CI that crosses zero. At this corpus size these are three clean null results. The scene-reference colour channel needs a caution. It nominally leads on PSNR, SAM and ERGAS among non-leaked models, but it is the *worst* model in the table on FID and LPIPS, including the leaked row. A flat, scene-averaged colour fed in as a second channel may produce a low-frequency block artefact that pixel-wise metrics reward and perceptual metrics penalise. We have not confirmed this visually, so we do not recommend the variant.

C. The Perception–Distortion Trade-off Repeats

The ResShift bridge again trades PSNR for FID on this corpus (179.9 against 311.0 for the regressor). This repeats the central result of Sec. IV on data the bridge has not seen, with a fifth of the training data. The trade-off belongs to the two architectures, not to one corpus.

TABLE VI
SCENE-DISJOINT TEST SPLIT OF THE SECOND CORPUS ($n = 1657$). PSNR IS THE POINT ESTIMATE ± THE HALF-WIDTH OF A SCENE-LEVEL BOOTSTRAP 95% CI. BEST AMONG NON-LEAKED MODELS IN BOLD. THE NOMINAL WINNER OF A COLUMN IS NOT ALWAYS THE RECOMMENDED MODEL (SEE TEXT).

Model	PSNR (dB)	SSIM ↑	LPIPS ↓	FID ↓	SAM° ↓	ERGAS ↓	CC ↑	Params (M)
Baseline (v3 recipe, this corpus)	13.16 ±1.14	0.129	0.829	311.0	7.24	61.27	0.110	859.6
Dual-SAR input	13.17 ±0.94	0.130	0.699	243.9	7.26	61.86	0.136	859.6
Wavelet low-frequency loss	13.33 ±0.98	0.135	0.721	252.5	7.46	60.49	0.151	859.6
Scene-reference colour†	**13.45**±0.61	0.121	1.015	365.7	**6.44**	**58.95**	0.094	859.6
ResShift bridge	12.76 ±0.75	0.086	0.766	**179.9**	8.65	64.39	0.049	859.6
Curated training data (3% dropped)	13.13 ±0.96	**0.136**	0.694	242.1	7.34	61.21	0.176	859.6
Perceptual LoRA fine-tune	11.89 ±0.89	0.072	0.709	303.2	9.92	70.72	0.006	859.6
Perceptual DoRA fine-tune	10.95 ±0.91	0.071	0.687	287.5	10.86	78.73	0.051	859.6
Perceptual MiCA fine-tune	11.83 ±0.65	0.041	0.712	305.6	9.52	71.81	0.049	859.6
Distilled student (half width/depth)	13.28 ±0.96	0.134	0.860	325.9	7.54	60.16	0.129	**148.1**
Pixel-space baseline	13.31 ±0.98	0.130	**0.638**	190.9‡	7.52	60.42	**0.215**	**49.9**
Per-patch split (diagnostic)	16.16 ±0.79	0.232	0.720	235.3	4.92	44.17	0.509	859.6

†Best nominal PSNR but worst FID and LPIPS in the table. ‡Second only to the bridge.
The per-patch row is a diagnostic below the rule, not a candidate model; see Sec. VIII-A.

D. Backbone Capacity Is Not the Binding Constraint, Again

Sec. IV-C showed that a *better* autoencoder gives a *worse* translator. Here a plain convolutional U-Net on 256 px RGB, with no autoencoder, no latent space and no diffusion sampling, matches the full pipeline's PSNR (overlapping 95% CIs). It has 49.9 million parameters against 859.6 million for the regressor, and it *improves* LPIPS, CC and FID. Its FID approaches the bridge's 179.9 in one forward pass instead of fifteen. Together with Sec. IV-C, this is a second, independent sign that neither autoencoder quality nor backbone capacity limits the task. The oracle decomposition (Sec. V) says what does: a per-scene radiometric offset that is large and not a function of the radar input. A small network cannot predict that offset better than a large one, and it matches the large one in fidelity while producing more coherent images.

E. Parameter-Efficient Fine-Tuning Did Not Close the Perceptual Gap

Motivated by the bridge's FID advantage, we fine-tuned low-rank adapters on the frozen baseline with an adversarial and LPIPS objective. We compared three initialisations: LoRA [20], DoRA [21] and MiCA [22], which starts the adapter from the *minor* (smallest) singular directions of each target weight. All three cost PSNR without recovering FID. The best on FID (DoRA, 287.5) is still far from the bridge's 179.9, and it costs 2.22 dB more PSNR than LoRA's own 1.28 dB cost. This is not evidence against any of the three methods. They were trained for 6000 steps with one untuned setting, against 15 000 steps for the core recipe. MiCA was validated by its authors on knowledge injection in language models, not on diffusion U-Net regression. A fair ranking needs a matched budget and at least a coarse learning-rate sweep per method.

F. Distillation: A Modest Efficiency Result

Distilling the baseline into a student with half the width and depth (148.1 million parameters, 17 % of the teacher) gives PSNR that is statistically indistinguishable from the full model (95% CI $[-0.17, +0.31]$ dB), with somewhat worse FID and LPIPS. This is an ordinary distillation trade-off, but a $6\times$ parameter cut at no measurable fidelity cost is useful for deployment.

G. Test-Time Techniques and Quantisation

On the same 1657-image test set, an eight-way dihedral test-time ensemble and a last/best weight soup each change PSNR by less than 0.05 dB, with overlapping intervals. Simulated INT8 weight-only quantisation of the regressor costs 0.01 dB, in line with reports that post-training weight quantisation is nearly free for diffusion-style U-Nets. We do not report the posterior-mean and perception–distortion blend sweep over bridge samples, or the fp32-versus-bf16 latency benchmark. Both were run on a fixed-position subset of the test list, and we later found that both architectures score over 2 dB lower on that subset than on the full test set. That is a property of the subset, but it makes the two analyses unsafe to read quantitatively until they are re-run on a random sample. Giving more weight to the bridge in the blend lowered FID steadily, in line with Sec. IV.

IX. POSTERIOR-MEAN RECTIFIED FLOW ON THE OFFICIAL SEN1-2 RELEASE

Sections V and VIII suggest two things. First, the residual is dominated by information the radar does not carry, so the best PSNR on unseen scenes is set by the posterior mean $\mathbb{E}[X \mid Y]$ and not by model capacity. Second, a sampler from the posterior pays about twice the MMSE, which matches the 1.6 dB gap between bridge and regressor in Table II. PMRF aims at the point between the two: the lowest MSE reachable by an estimator whose outputs follow the optical distribution [27]. We test it on new data with a stricter protocol.

A. Data and Protocol

We stream the official SEN1-2 release [15] (282 384 pairs, four seasonal ROI sets) from its archive and keep a deterministic, hash-selected subset of whole scenes. It is split 80/10/10 by *scene* within each season. SEN1-2 tiles each scene with a stride of 128 pixels, so neighbouring patches share bit-identical

halves. We use this to rebuild each scene mosaic from SAR alone, and attach to every patch a 512^2 SAR context window downsampled to 128^2. The context uses no optical data, so it is valid at test time. Metrics follow Sec. III: per-image PSNR on 8-bit RGB, averaged.

B. Method

Stage A (Fig. 6) is a 53.5 M-parameter pixel-space U-Net f_ω. It is conditioned on the season by FiLM and on the context by a convolutional encoder, whose global vector modulates every block and whose map is added at the bottleneck. It is trained with pure MSE, $\omega^* = \arg\min_\omega \mathbb{E}\|X - f_\omega(Y)\|_2^2$, so that $\hat{x}^* = f_{\omega^*}(Y)$ approximates $\mathbb{E}[X \mid Y]$. Perceptual or SSIM terms would pull it away from the posterior mean that PMRF needs.

Stage B is a 92 M-parameter U-Net g_θ that moves $Z_0 = \hat{x}^* + \sigma_s\epsilon$, with $\epsilon \sim \mathcal{N}(0, I)$ and $\sigma_s = 0.1$, to $Z_1 = X$ along $Z_t = tX + (1-t)Z_0$. In pixel space it predicts the clean image, not the velocity, and minimises the velocity-equivalent loss

$$\mathcal{L}_B = \mathbb{E}_{t,X,\epsilon} \frac{\|g_\theta([Z_t, \hat{x}^*, Y], t) - X\|_2^2}{\max(1-t, 0.05)^2}, \tag{4}$$

with t drawn from an equal mix of logit-normal and uniform. Sampling uses K Euler steps $Z_{t+\Delta} = Z_t + \frac{\Delta}{1-t}(g_\theta - Z_t)$, which land exactly on the predicted image at $t = 1$, so $K = 1$ is a valid one-step sampler. Both stages use AdamW, batch 32, cosine decay, EMA 0.9999 and bf16 on one A100 (3.3 and 2.5 iterations/s).

C. Results

Fig. 7 shows the runs. Stage A reaches its best validation PSNR of 12.94 dB at step 10 000, then drifts to 12.74 dB by step 25 000 while its training MSE keeps falling (0.143 to 0.115). The model keeps fitting the training scenes but stops generalising to new ones. This is what the oracle decomposition predicts. The plateau lies within the confidence interval of the scene-disjoint baseline in Table VI (13.16 dB). It is a further sign that SEN1-2 fidelity on unseen scenes is near 13 dB, and that the 18 dB to 20 dB reported under per-patch splits, including our own Table II, mostly reflects the split.

The PMRF flow, trained on the cached posterior means, improves steadily from 12.16 dB at step 5000 to 12.53 dB at step 35 000, with gains down to 0.004 dB per 5000 steps, against 12.69 dB for its own input. A cost of 0.16 dB is an order of magnitude below the 1.6 dB that posterior sampling cost the bridge, as the PMRF guarantee suggests. PSNR cannot show whether this small cost buys the perceptual gain that motivates PMRF. We did not run a perceptual evaluation of this pixel-space flow, so we make no claim about it.

X. Continued Training and Latent Flow Refinement

The error budget of Sec. V says the remaining error of regressor v3 is mostly low-frequency radiometry. We tested two ways of moving v3 along the distortion–perception curve.

TABLE VII
Continued training and latent flow refinement (4096 test patches, one draw per stochastic method). Best per column in bold. Published rows are quoted from the original papers and may use a different protocol.

Method	PSNR↑	SSIM↑	LPIPS↓	FID↓	SAM↓	CC↑
v3 (regressor)	**16.60**	**0.253**	0.706	142.5	**7.26**	**0.372**
v3$^+$ (continued)	15.19	0.192	0.703	132.8	8.04	0.342
PMRF-latent, NFE 8	14.79	0.159	0.667	208.7	8.34	0.276
PMRF-latent, NFE 16	14.44	0.145	0.673	205.5	8.53	0.250
Blend λ=0.25	15.15	0.188	0.664	**130.3**	8.08	0.330
Blend λ=0.50	15.01	0.176	**0.651**	153.6	8.15	0.305
Blend λ=0.75	14.77	0.161	0.660	181.6	8.29	0.276
DDPM [25] (quoted)	19.72	0.312	n/a	116.9	n/a	n/a
ACD, 16 steps [26] (quoted)	20.57	0.323	n/a	116.9	n/a	n/a

Both start from its weights and use the same 16-channel latent space, conditioning and losses (Sec. III).

(i) Continued training (v3$^+$). v3 is trained further with its original recipe (shift-tolerant latent loss, pixel L1, SSIM, MIND and SAM terms, decoder learning rate 5×10^{-6}, EMA 0.9995) at an effective batch of 8 and a learning rate of 1×10^{-5}. We set a paired gate on a fixed monitor set before the run: continue only if the paired PSNR gain over v3 at 20 000 steps is at least 0.15 dB.

(ii) Latent posterior-mean rectified flow (PMRF-latent). Following PMRF, a second U-Net, initialised from v3 with three input groups $[x_t, y, \hat{z}]$, regresses the velocity $z_0 - x_0$ of a straight path from $x_0 = \hat{z} + \sigma\varepsilon$ (σ=0.1) to the target latent z_0. Here $\hat{z}$ is the regressor output and y the SAR latent. Sampling uses Euler steps (NFE $\in \{8, 16\}$). We also report latent blends $\hat{z} + \lambda(z_{\text{flow}} - \hat{z})$. The flow was gated at 10 000 of a planned 40 000 steps: continue only if PSNR falls by at most 2 dB *and* LPIPS falls by at least 0.03.

Protocol. All rows are scored on the same 4096 patches from the test scenes of the official-release subset of Sec. IX, with the metric definitions of Sec. III, FID [16] and LPIPS [17]. The official release has no terrain label, so the text condition is the empty prompt. It scored higher than a season-plus-terrain prompt for v3 on the validation scenes (16.02 dB against 15.53 dB). v3 was originally trained on a per-patch split, so absolute values are not comparable with the scene-disjoint results of Secs. VIII and IX. The comparison between the rows of Table VII is valid.

Results. Neither experiment improves fidelity. Continued training lowered PSNR by 1.41 dB (paired, $\pm$0.05, n=4096) and SSIM by 0.061. Its monitor-set curve was already 1.1 dB below v3 at 20 000 steps while its training loss kept falling, so the gate stopped it. It did lower FID from 142.5 to 132.8, so it moves along the distortion–perception curve without improving it. The latent flow lowered LPIPS from 0.706 to 0.667 at NFE 8, but raised FID to 208.7 and cost 1.8 dB. Its gate stopped it at 10 000 of 40 000 planned steps, so we do not claim it has converged. The blends show where the flow helps and where it hurts. The lowest FID (130.3) is at λ=0.25 and FID grows with λ, so the texture the flow adds is not placed where the optical image has it (Fig. 8). None of these variants comes near the quoted PSNR or FID of the published systems, and we make no state-of-the-art claim for them.

Colour. Fig. 8 shows what the metrics summarise. Every

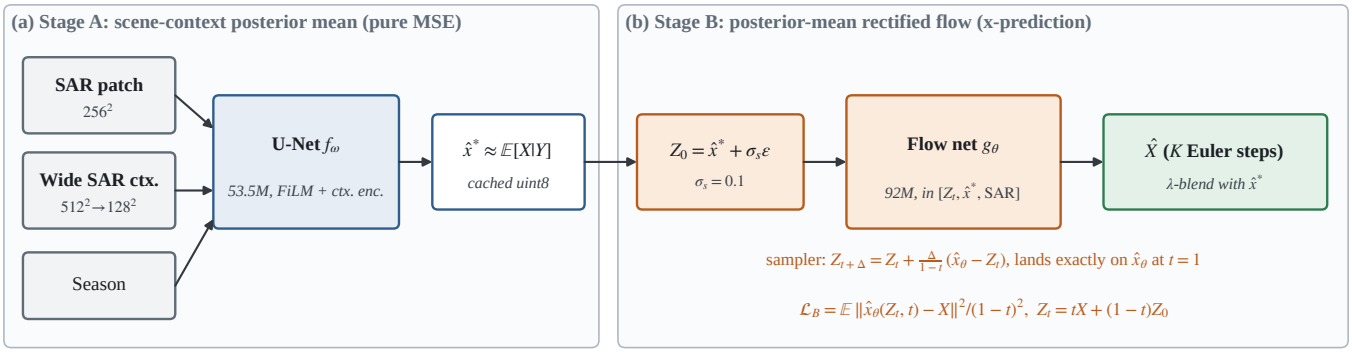


Fig. 6. PMRF pipeline. Stage A estimates the posterior mean from the SAR patch, its wide SAR context and the season, using a pure MSE loss. Its output is cached once. Stage B learns a rectified flow from the noised posterior mean to the optical image with x-prediction. K Euler steps give the output, and a convex blend with $\hat{x}^*$ traces the distortion–perception curve at no extra cost.

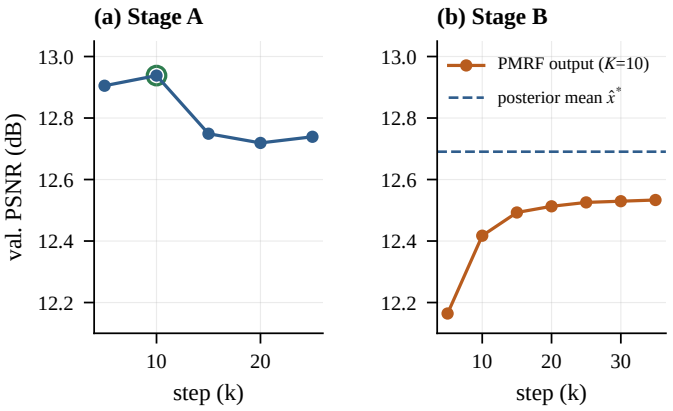


Fig. 7. Validation PSNR on held-out scenes (fixed 512-image subset, EMA weights). (a) Stage A peaks at step 10k (circle) while the training loss keeps falling. (b) The PMRF output ($K = 10$) rises from 12.16 dB to 12.53 dB and ends 0.16 dB below the cached posterior mean (dashed).

variant renders the scenes in a dark, desaturated brown palette. The spring ground truth is green, and all three predictions are brown. On the four tiles shown, v3$^+$ has the higher PSNR in two (spring and winter), ties in summer and is lower in fall, while over the 4096 test patches it is 1.41 dB lower (Table VII). PMRF-latent adds texture and some colour variation but has lower PSNR than v3 in three of the four rows. This is the regression-to-the-mean behaviour expected when the radar image does not determine colour (Sec. V), and it agrees with Aydin et al., who find that SAR alone does not carry enough information to reproduce the optical scene [24]. Colour supervision [25] and colour-attention diffusion [23] target the same problem on SEN1-2, so any colour-specific component must be compared with them directly.

XI. Limitations

The comparison with published numbers is not protocol-matched. Our validation set is the full 15 394-image stratified split, and other work may evaluate on curated subsets. Converting Sentinel-2 reflectance to 8-bit RGB also involves a stretch choice (fixed divisor, percentile or histogram equalisation), and a more compressed dynamic range raises PSNR mechanically. We could not verify the rendering used by the reference result, so part of the 1.40 dB difference may come from protocol and not from method. Cross-paper absolute comparisons on this benchmark should be read with caution. Our within-paper comparisons all use one harness.

Our train and validation split is stratified per patch. SEN1-2 patches are cut from larger scenes with overlap, so a per-scene split would be stricter. Sec. VIII-A audits this on an independent corpus and finds a 3.00 dB PSNR inflation (95% CI $[2.06, 4.04]$) from per-patch over per-scene splitting. We have not repeated that audit on the split behind Table II, so we can give the direction and a plausible size of the optimism but not its exact value for the headline numbers.

The oracle decomposition uses $n = 400$ images and the calibration analysis uses 150-image splits. Both would benefit from the full set. The ERGAS implementation error noted in an earlier draft (a unit mismatch between numerator and denominator that doubled the metric) has been corrected. Table VI reports the corrected value, but we have not recomputed it for Table II. The PMRF study of Sec. IX reports validation curves only, with no test-scene FID or LPIPS and no flow-from-noise control that isolates the contribution of PMRF. LPIPS, FID and ERGAS were never recorded for regressors v1 and v2 or the FLUX-latent regressor in Table II, and cannot be rebuilt from the released files, so they are left as n/a. Stage A fits its training scenes better than unseen ones, so Stage B is trained on posterior means that are more accurate than the ones it sees at test time. Out-of-fold caching would remove this known weakness of two-stage pipelines.

XII. Conclusion

We presented a SAR-to-optical system that reaches 18.32 dB PSNR and 0.306 SSIM on the SEN1-2 validation split from a single GPU, together with a stochastic branch that holds the best perceptual score at FID 128.2. The technical enabler is a diffusability-aware autoencoder adaptation. The practical surprise is that a single-pass deterministic regressor beats an iterative diffusion bridge on every fidelity metric.

Our more lasting contribution is diagnostic. The oracle decomposition shows that the residual error is low-frequency and not structural, that misregistration is worth only 0.14 dB,

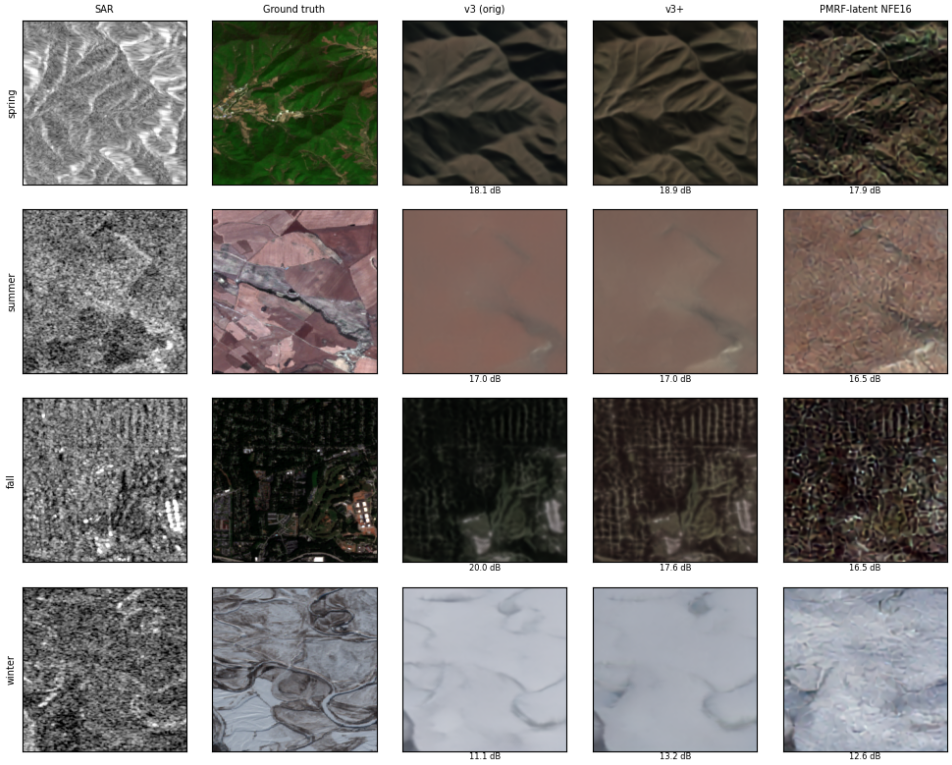


Fig. 8. One test patch per season (rows: spring, summer, fall, winter). Columns: SAR input, Sentinel-2 ground truth, v3, v3^{+} and PMRF-latent (NFE 16). The number under each prediction is the PSNR of that tile.

and that the largest recoverable part, a per-scene radiometric offset worth up to $2.27\,\mathrm{dB}$, cannot be predicted from the radar input. A much better autoencoder gave a worse translator. Effort should therefore go not into larger backbones or better latents but into supplying information the radar image does not contain, such as multi-temporal stacks, auxiliary geospatial layers or acquisition-matched data.

The scene-disjoint ablation (Sec. VIII) supports this in two ways. It measures the patch-level split leakage at $3.00\,\mathrm{dB}$ PSNR, and it shows that a 49.9-million-parameter plain convolutional network, with no autoencoder, no latent diffusion and a single forward pass, matches our 859.6-million-parameter pipeline in fidelity while improving every perceptual metric we measure. We tested two refinements of the regressor, continued training and a latent PMRF head (Sec. X). Neither improved fidelity, and both left the colour problem unsolved. Colour is the open problem: a method that supplies it must do so without using the ground truth, and must be compared with existing colour-supervised diffusion models. We release all checkpoints, the evaluation code and the oracle suite, together with the second corpus and its ablation code. Pixel/semantic dual-LoRA fine-tuning [29] is another direction, since it exposes the distortion–perception trade-off as an inference-time control.

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